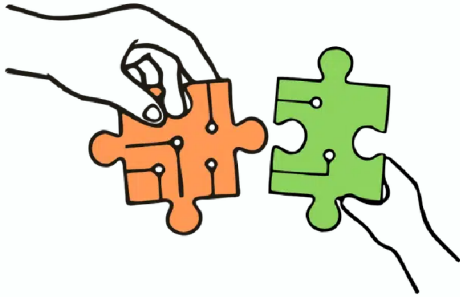


Child-Centred AI workshop at CHI2026



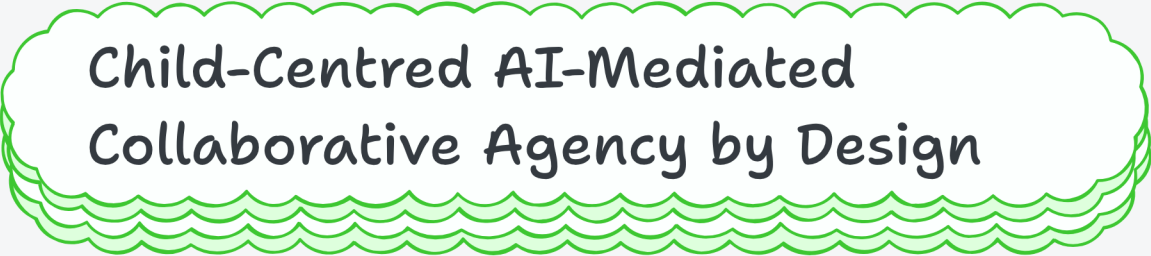
Presented by Vidminas Vizgirda
April 2026

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by Vidminas Vizgirda, Isobel Voysey, Zaki Pauzi, Najme Babai, Eva
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Child-Centred AI-Mediated
Collaborative Agency by Design

Child-Centred AI-Mediated Collaborative Agency by Design

- Vidminas Vizgirda (postdoc @1,2)
- Isobel Voysey (postdoc @1)
- Zaki Pauzi (postdoc @2)
- Najme Babai (PhD student @3)
- Eva Durall Gazulla (postdoc @3)
- Jane Waite (senior researcher @4)
- Ayça Atabey (postdoc @5)
- Sarah Turner (research fellow @2)
- Manolis Mavrikis (professor @2)
- Jun Zhao (senior researcher @1)

@1 = University of Oxford
@2 = University College London
@3 = University of Oulu
@4 = Raspberry Pi Foundation
@5 = University of Edinburgh

Supporting grants:

- CHAILD - Children's Agency In the age of AI: Leveraging InterDisciplinarity (MR/Z505882/1)
- GRASPING DATA: Co-creating Physicalizations to Empower Young Children to Interact with, Understand, and Benefit from Their Personal Data (MR/Z505602/1)
- Critical DataLit: Cultivating justice-oriented data literacies among GenZ (Grant #354445)

Workshop website

`oxfordhcc.github.io/CAMCAD/`



What is "agency"?

Agency in workshop submissions

Enacting Collaborative Agency through Critical AI Literacy: A Community-Based Program for Black and Brown Middle School Girls and Their Families
 Maria Kaban - mkaban@umd.edu - College of Information, University of Maryland, College Park
 Shirena Green - sgreen@umd.edu - College of Information, University of Maryland, College Park

Problem Space
 Young women increasingly shape online life at education, living, politics, healthcare, and media representation [1]. Yet Black and brown communities are disproportionately impacted by algorithmic bias, surveillance, data extraction, and exclusion from technological decision-making (D&M). At the same time, these communities remain underrepresented in the design, governance, and policy conversations that shape how AI systems are built and deployed [2]. Youth, particularly girls of color, are even more rarely positioned as legitimate contributors to conversations about AI's social impact [3]. An education effort focuses on technical skills or passive awareness rather than cultivating the capacity to question, critique, and influence sociotechnical systems, in response, we see *How can critical AI literacy programs intentionally structure collaborative agency across youth, families, and educators, so that young people move from learning about AI to selectively shaping conversations about the ethical and social use?*

Setting
 This study draws on a two-term, community-based informal learning program implemented in August 2024 in partnership with a local nonprofit organization focused on youth and family capacity building in underserved communities. The program was designed as a critical AI literacy experience integrating technical, sociotechnical critique, and creative advocacy. The program included daily guest speakers-local artists, UK engineers, advocates, public servants, and technologists-who addressed technical, sociotechnical, and policy issues, and local community site visits exposed youth to research labs and innovation centers, as well as real-world STEM pathways.

Figure 1. A circular diagram showing the relationship between 'This is the Best' critical AI literacy program and various components: Technical Skills, Sociotechnical Critique, Creative Advocacy, and Community Engagement. The diagram is divided into four quadrants, each with a specific focus: Technical Skills (AI Literacy, AI Literacy, AI Literacy), Sociotechnical Critique (AI Literacy, AI Literacy, AI Literacy), Creative Advocacy (AI Literacy, AI Literacy, AI Literacy), and Community Engagement (AI Literacy, AI Literacy, AI Literacy).

PDF

critically examine AI systems, make informed decisions, and enact decisions

control own actions and external environment

Trust Game

Rule: decide how many of four gold coins to give to the agent, knowing that the coins would be tripled and that the agent might return some, all, or none.

Measure: Children's sense of agency was measured through (1) willingness to give coins, (2) total coins given, and (3) the expected number of coins returned.

Anthropomorphic cues

PDF

Gather to Glow: Designing an Autonomy-Supportive Game for Parent-Child Digital Co-Play

ABSTRACT
 Parent-child digital co-play offers a challenge to design in which caregivers' involvement can unintentionally undermine children's autonomy, potentially undermining children's agency. Drawing on our co-designer theory (CDT), we present Gather to Glow, an autonomy-supportive game that integrates a digital co-play experience with a social support network. The game is designed to support children's autonomy by ensuring caregivers' autonomy through three cues: (1) social support cues, (2) digital support cues, and (3) social support cues. The game is designed to support children's autonomy by ensuring caregivers' autonomy through three cues: (1) social support cues, (2) digital support cues, and (3) social support cues.

initiate, interpret, and shape activity

PDF

Learning through Conversational Games with AI and a Robot

METHODS
 A longitudinal study with 10 children and 10 parents over 12 weeks. The study was designed to explore the impact of conversational games on children's learning and engagement.

GAMES
 The study used two types of games: a text-based game and a robot-based game. The text-based game was designed to be played with a parent, while the robot-based game was designed to be played with a robot.

EXAMPLE
 The study used a conversational game called 'The Robot's Secret' to explore the impact of conversational games on children's learning and engagement.

ETHICS
 The study was approved by the Institutional Review Board (IRB) at the University of Maryland, College Park.

CONCLUSION
 The study found that conversational games with AI and a robot can improve children's learning and engagement.

SOURCES
 The study was supported by the National Science Foundation (NSF) grant #1008308.

PDF

take an active part in research and product development

generate ideas, reflect, justify them, revise them, and act on them

Designing Minds in Dialogue: Centering AI as a Scaffold for Children's Cognitive Agency

RESEARCH QUESTION
 How can AI be designed to support children's cognitive agency in a way that is both effective and ethical?

WHY EARLY CHILDHOOD MATTERS
 Early childhood is a critical period for the development of cognitive agency, and AI can play a role in supporting this development.

THE 'MIND' CONSTRUCTION
 The 'Mind' construction is a framework for understanding how AI can be designed to support children's cognitive agency.

2 CHALLENGES
 The study identified two challenges: (1) how to design AI to support children's cognitive agency, and (2) how to ensure that the design is ethical.

PDF

make choices and decisions free from undue external influence

AI-Mediated Scaffolding for Student Agency and Peer Collaboration

INTRODUCTION
 In early childhood, agency refers to the capacity to influence one's own actions and decisions. This capacity is foundational for the development of autonomy, self-efficacy, and purpose (Vygotsky, 1978; Bandura, 2002). Fostering this capacity has proven a complex task, as it requires the cultivation of a supportive environment that encourages exploration, risk-taking, and the development of problem-solving skills.

CONCLUSION
 The study found that AI-mediated scaffolding can effectively support student agency and peer collaboration.

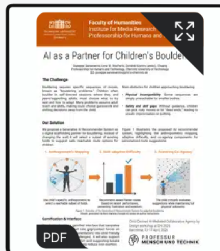
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What is "collaborative agency"?

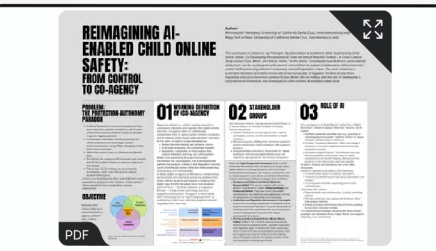
Co-agency in workshop submissions



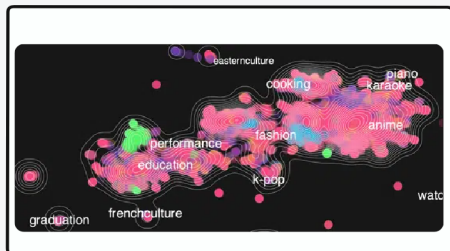
Shared decision making and active engagement
Example: collaborative learning with peers and AI



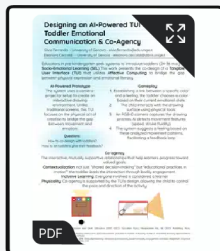
Asymmetric, role-differentiated partnership
Example: AI recommender system for bouldering



Joint understanding, influence, and regulation
Example: shared online safety decisions



Working with others on a shared problem
Example: tackling algorithmic profiling opacity



Interactive, mutually supportive relationships
that help learners progress towards goals



Shared production of action and meaning
Example: child-robot interaction

"Collaborative agency" by design?

Workshop Agenda

1

14:15-15:45 Invited presentations

2

15:45-16:30 Poster session

3

16:30-18:00 Break-out activities

After the workshop: IJCCI SI on "Agency in Child-AI Interaction"

- Special Issue in the International Journal of Child-Computer Interaction
- Invites full papers agency in child-AI interaction
- Call for papers out now ->
- Open for submissions for 6 months
 - Open: 27 April 2026
 - Mid-feedback: 27 July 2026
 - Close: 27 October 2026



[sciencedirect.com/special-issue/331176/
agency-in-child-ai-interaction](https://sciencedirect.com/special-issue/331176/agency-in-child-ai-interaction)

Join the CCAI 2026 community
discord.gg/6kuHZdetEJ

